

Tutorial No.11

Q.1. Determine whether following differential equations are exact or not? If yes find a solution

- 1) $(3x + 2y) dx + (2x + y) dy = 0.$
- 2) $(y^2 + 3) dx + (2xy - 4) dy = 0.$
- 3) $(3x^2 y + 2) dx - (x^3 + y) dy = 0.$
- 4) $(6xy + 2y^2 - 5) dx + (3x^2 + 4xy - 6) dy = 0$
- 5) $(y \sec^2 x + \sec x \tan x) dx + (\tan x + 2y) dy = 0$

Q.2. Solve following differential equations

- 1) $\frac{dy}{dx} + \frac{y}{x} = x^3 - 3$
- 2) $\frac{dy}{dx} + y \cot x = \cos x$
- 3) $\frac{dy}{dx} + y \sec x = \tan x$
- 4) $x \cos x \frac{dy}{dx} + y(x \sin x + \cos x) = 1$
- 5) $dr + (2r \cot \theta + \sin 2\theta) d\theta = 0$
- 6) $\frac{dy}{dx} + y \tan x = \cos x$
- 7) $x \log x \frac{dy}{dx} + y = 2 \log x$

Q.3 For each of the DE mentioned below 1) Find complementary function

2) Find particular integral by method of undetermined coefficients

- 1) $(D^2 + 1)y = x \sin x$
- 2) $(D^2 - 4D + 3)y = x^3 e^{2x}$
- 3) $(D^3 + 4D)y = \sin 5x \cos 3x$
- 4) $(D^2 - 4D + 4)y = x e^{2x} \sin x$
- 5) $(D^2 + 5D + 4)y = x^2 + 7x + 9$
- 6) $(D^2 - 2D + 1)y = x e^x \sin 2x$
- 7) $(D^3 - 4D^2 + 4D)y = e^{2x} + e^x + 1$
- 8) $(D^2 + 3D + 3)y = 2 \sin 4x$
- 9) $(D^3 + 3D^2 + 3D + 1)y = e^{-x}$

Q.4 For each of the DE mentioned below 1) Find complementary function

2) Find particular integral by method of variation of parameters

- 1) $(D^2 + 4)y = \sec^2 2x$
- 2) $(D^2 - 6D + 9)y = x^{-2} e^{3x}$
- 3) $(D^2 + 1)y = x + \tan x$
- 4) $(D^2 + 2D + 1)y = e^{-x} \log x$
- 5) $(D^2 - 4D + 4)y = e^{2x} \sec^2 x$
- 6) $(D^2 - 2D + 2)y = e^x \tan x$
- 7) $(D^2 + 1)y = \cot x$
- 8) $(D^2 + 3D + 2)y = e^x$
- 9) $(D^2 - 1)y = \frac{2}{1 + e^x}$